Qualcomm Technologies, Inc.

Qualcomm® Snapdragon™ 410E (APQ8016E) Windows Display Overview

LM80-P0337-3 Rev. B

Qualcomm Technologies, Inc.
5775 Morehouse Drive
San Diego, CA 92121
U.S.A.

Revision History

Revision	Date	Description
A	January 2018	Initial release
B	January 2018	Updated the document as per the branding changes

Contents

Section 1

Introduction

Objective

This document is intended for customers who are familiar with APQ8016E windows display, including:

- Distinction between display-related hardware and software components
- Display capabilities and performance benefits of the Multimedia Display Subsystem (MDSS)
- Control flow and data flow
- Basic information on source code layout, build, and debugging

Acronyms, abbreviations, and terms

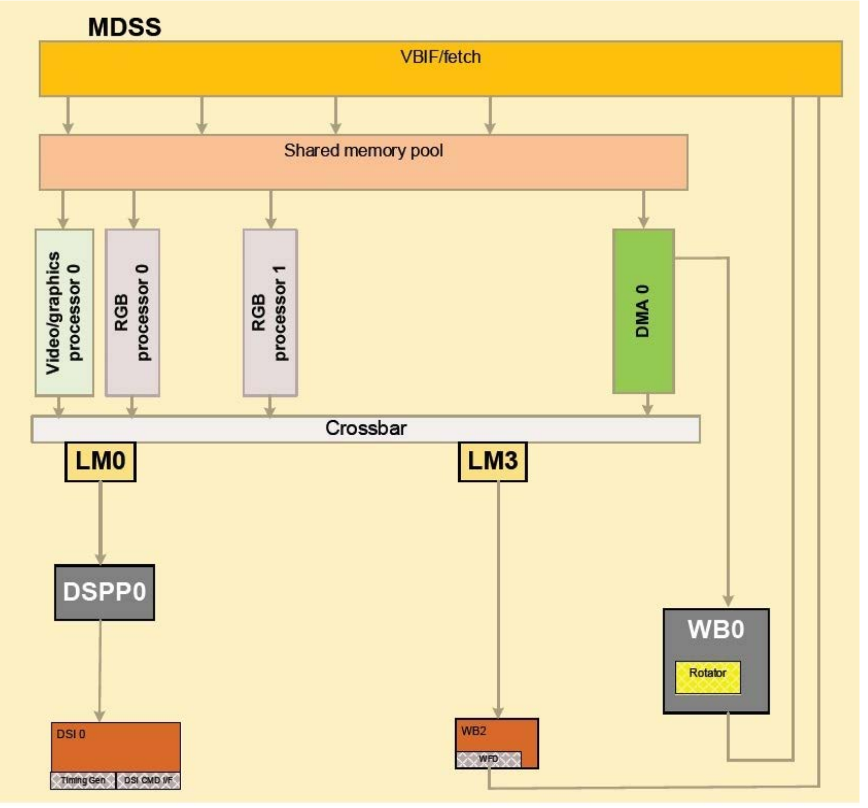
Term	Definition
ACPI	Advanced configuration and power interface
CABL	Content Adaptive Backlight Leveling
CAF	Content adaptive filter
CDD	Canonical Display Driver
DSPP	Destination surface processor
DSI	Display serial interface
GOP	Graphics Output Protocol
LM	Layer mix
MDSS	Multimedia Display Subsystem
SSPP	Source surface processor pipes
SBC	Smooth Backlight Control
WB	Write-back

Section 2

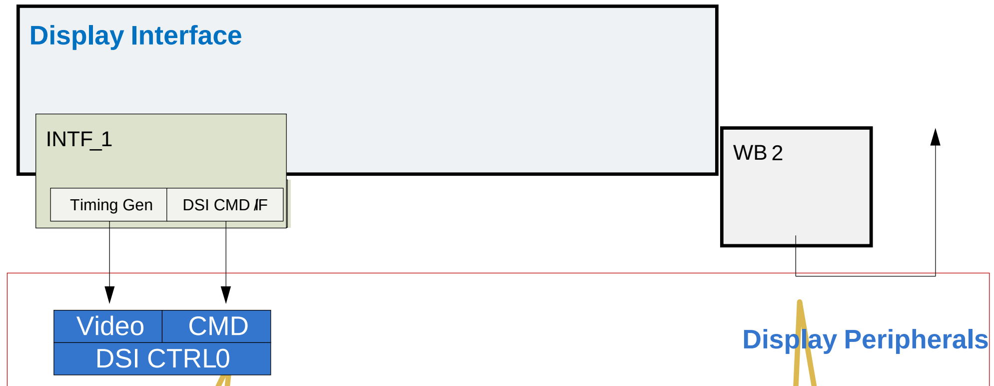
System Architecture

MDSS Overview

- Source surface processor (ViG, RGB, DMA – SSPP)
  - Format conversation and quality improvement for s
    surfaces (video, graphics, and so on)
- Layer mixer (LM)
  - Blend and mix source surface together
- Destination surface processor (DSPP)
  - Conversation, correction, and adjustment based on the panel characteristics
- Write-back/rotation (WB)
  - Write back to memory
  - Perform rotation if necessary
- Display interface
  - Timing generator and interface connection the display peripheral



Display Peripheral



MIPI DSI

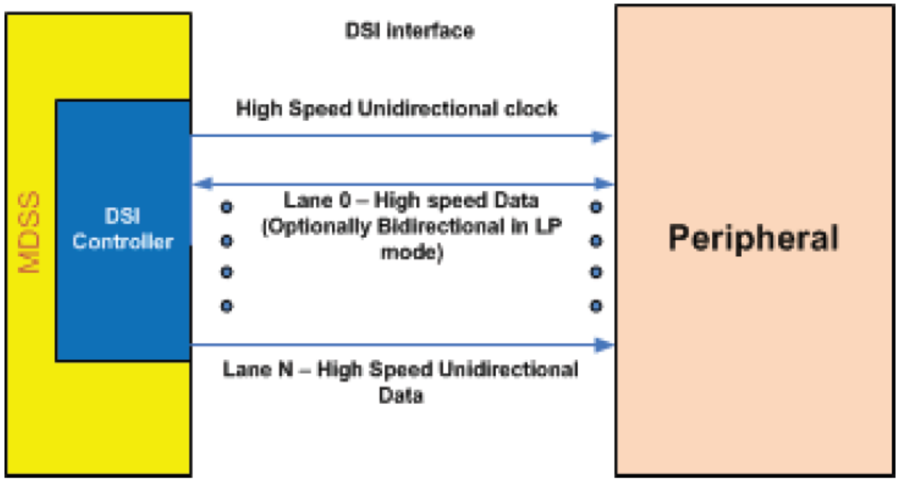
- One MIPI DSI
- Supports both smart and dumb displays
- Up to four lanes support for each DSI
- Up to 1.5 Gbps per lane with 24 bpp support
- DATA±, CLK± (for one lane)
- Supports up to 1280 × 800 resolution with 60 fps

WFD

- 1280 × 720 at 30 fps

MIPI Display Serial Interface (DSI)

- The DSI controller is implemented to support the MIPI alliance standard for DSI.
- The DSI controller includes one high-speed clock lane and one or more data lanes. Each lane is carried on two wires and uses low voltage differential signaling.
- There are two modes of operations for DSI-compliant peripherals:
  - Command mode
  - Video mode



Section 3

Feature Overview

Supported Interfaces

- APQ8016E supports up to two concurrent displays
  - DSI
    - Up to 1200 × 800 at 60 fps
  - Wi-Fi display
    - 1920 × 1080 at 60 fps

Note: The resolutions listed here are the maximum resolutions for an individual interface. When driving three displays simultaneously, it may be necessary to reduce the size of the connected displays. The end-to-end performance numbers might be different than the above.

Source Surface Processor Pipes (SSPP)

Feature	ViG	RGB	DMA
Number of pipes	1	2	1
	• 24-bit RGB (888) • 16-bit RGB (565) • 16-bit x/ARGB (4444,1555) • 32-bit x/ARGB (8888) (with ARGB/RGBA/ABGR/BGRA and RGB/BGR premutation) • YCbCr422 interleaved (YCrYCb, YCbYCr, CbYCrY, and CrYCbY) • AYCrCb444 interleaved • YCbCr420 pseudo planar (NV12 and NV21) • YCbCr422 pseudo planar (H1V2 and H2V1) • NV12/NV21 + alpha • YCbCr422 pseudo planner + alpha • YCbCr420 planar • YCbCr422 planar	• 24-bit RGB (888) • 16-bit RGB (565) • 16-bit x/ARGB (4444,1555) • 32-bit x/ARGB (8888) (with ARGB/RGBA/ABGR/BGRA and RGB/BGR premutation)	• 24-bit RGB (888) • 16-bit RGB (565) • 16-bit x/ARGB (4444,1555) • 32-bit x/ARGB (8888) (with ARGB/RGBA/ABGR/BGRA and RGB/BGR premutation) • YCbCr422 interleaved (YCrYCb, YCbYCr, CbYCrY, and CrYCbY) • AYCrCb444 interleaved • YCbCr420 pseudo planar (NV12 and NV21) • YCbCr422 pseudo planar (H1V2 and H2V1) • NV12/NV21 + alpha • YCbCr422 pseudo planner + alpha • YCbCr420 planar • YCbCr422 planar
Scaling ratio	Arbitrary 1/64-20x (decimation when < 1/4)	No	No
Sharpening	Yes	No	No
Chroma up	Yes, any chroma sites	No	No
CSC	Yes	No	No
Content adaptive contrast enhancement	• 256-bin histogram • 256-entry LUT	No	No
Flip	Vertical and horizontal flip		

Layer Mixer, Background Color, and Hardware Cursor

Feature	APQ8016E MDP support
Number of layer mixers	Two
Maximum number of surfaces blended	Four + hardware cursor + background color
Total number of pipes for blending	10 (1 ViG + 2 RGB + 1 DMA)
Alpha blending	Constant alpha, per pixel alpha, premultiplied alpha, modulation alpha – Reverse alpha for all the above
Alpha blending for background color	Yes
Background color generation	Yes (no data fetch for background color)
Transparency color key	Source color key, destination color key, simultaneous source and destination color key, and color key range
Arbitrary blending order	Yes
Blending in linear space	Yes
Blending color depth	12 bits/component
Hardware cursor size	128 × 128

Destination Surface Processor Pipes (DSPP)

Feature	APQ8016E MDP support
Sunlight visibility improvement (SVI)	Yes
CABL	Yes
Panel color correction (PCC)	3 × 11 polynomial
Bit-depth for color correction	12 bits/component
Gamma correction	Yes (Three-channel LUT)
Picture adjustment (hue, saturation, contrast, and intensity)	In HSV space, smooth curve soft clip
Dither	4 × 4 ordered dithering performed without panel depth reduction
Memory color (sky, foliage, and skin tone)	Yes
Six-color zone adjustment	Yes

Rotator and WB

Feature	APQ8016E MDP support
Rotator	
Input format support	Same as ViG
Rotation modes	90º, 180º, and 270º
WB	
Number of WBs	Two, WB0 and WB2
WB performance	WB0 – 1280 × 800 @ 60 fps and WB2 – 1280 × 720 @ 30 fps
WB format	• 24-bit RGB (888) • 16-bit RGB (565) • 16-bit x/ARGB (4444, 1555) • 32-bit x/ARGB (8888) – With ARGB/RGBA/ABGR/BGRA and RGB/BGR permutation • YCbCr420 pseudo planar (NV12, NV21) • YCbCr422 pseudo planar (H1V2, H2V1) • NV12 • YCbCr422 pseudo planar • YCbCr420 planar • YCbCr422 planar

Wireless Display

Feature	APQ8016E MDP support
Number of WBs	One, WB2 with display processing
WB performance	WB2 – 720p at 30 fps
WB format	• 24 bit RGB (888) • 16 bit RGB (565) • 32bit x/ARGB or BGRx/A (8888) • YCbCr420 pseudo planar (NV12)
Composition	WB for the final composition surface and WB2 with hardware cursor

Section 4

Display and Video Processing Features

Scaling

Feature		ViG pipe	RGB pipe
Scaling	Scaling ratio	1/64-20x (decimation for <1/4)	No
	Upscaling filter	• 4-tap CAF (32 phases) • 2-tap bilinear (32 phases) • Nearest neighbor (32 phases)	No
	Downscaling filter	PCMN (8-phase)	No
Chroma up		• 420→444 and 422→444 • Combined with scaler • Any chroma sites by programming initial phase	No
• CAF – Content adaptive filter; adjust filter coefficients based on content • PCMN – Phase control M/N; fractional averaging filter for downscaling; fractional is done with the 8-phase PCMN, for example, uses 3.375 (27/8) averaging filter for 3.375:1 downscaling			

2x by 4 × 2

2x by QSEED2

48/100 AVG

48/100 QSEED2

Sharpening

Feature	ViG pipe	RGB pipe
Sharpening/smoothing	• -256 to 255 – Negative means smoothing and positive means sharpening • Combined with scaling filter	N/A

Before

After

Noise Injection/Dithering

- Reduces banding artifacts by adding random noise
- Two types of banding artifacts, two types of dithering
  - Destination dithering – Reduces banding artifacts due to insufficient color depth of the panel; for example, 24 bpp content on 18 bpp panel
  - Source dithering – Reduces banding artifacts in content even though both content and panel are true colors

Adaptive Backlight – Global CABL

Feature	APQ8016E MDSS
Process space	V (maximum (R,G,B)) of the HSV space
Histogram collection	Hardware, 256 bins
LUT	Hardware, 256 bins
Pipeline location	After blending and picture adjustment
Core algorithm	Software

Original image

Original image with CABL active

- Better visual quality with separated LUT for gamma correction and ABL
- Compensates impact of panel characteristic change due to backlight level variation
- Backlight savings of up to 55% possible but depends on the final OEM configuration

Sunlight Visibility Improvement (SVI)

SVI is a Qualcomm® Technologies, Inc. (QTI) proprietary global tone adjustment solution to improve the display visibility in bright, ambient light environments without boosting panel backlight/brightness.

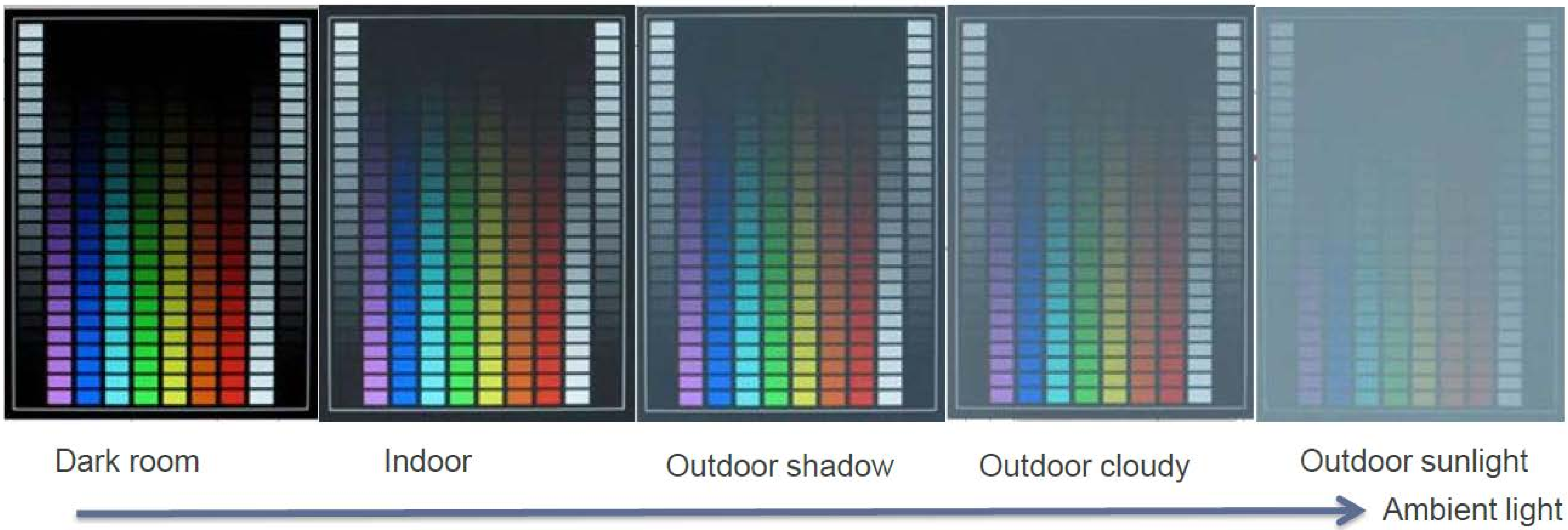
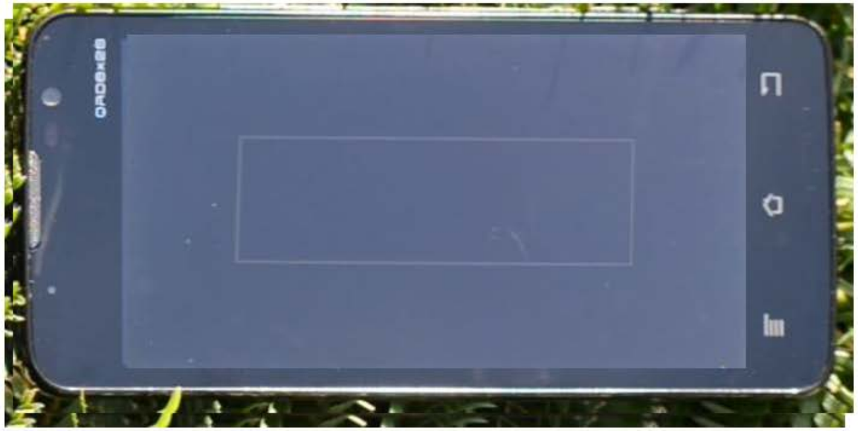


Figure: Camera shot taken on commercial device under different ambient lights, from left to right, with ambient light value 500 lux, 1000 lux, 2000 lux, 5000 lux, and 10000 lux.

Actual Example



without SVI, 100% BL

Extremely strong ambient light (33,400 lux)

with SVI, 100% BL

Section 5

Picture Adjustment

Picture Adjustment (HSIC)

Hue

Saturation

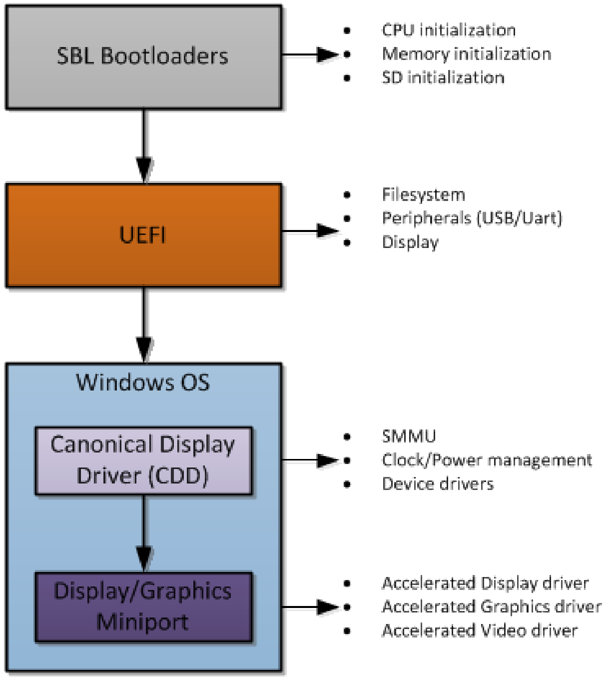
Brightness

Section 6

Windows on Snapdragon (WoS) Unified Extensible Firmware Interface (UEFI)

Display – Display Init Sequencing

- Platform boot consists of three stages:
  - SBL boot loaders, UEFI, and the main operating system.
  - The first two stages are non persistent; execution code is lost at the next stage.
  - The only persistent code between UEFI and the operating system is Advanced configuration and power interface (ACPI).
- Windows has two display driver modes:
  - Canonical Display Driver (CDD)
    - Used during bootup, but also used when the miniport is not loaded or upgraded.
  - Display/graphics miniport driver
    - Provides accelerated display, graphics, and video functionality.



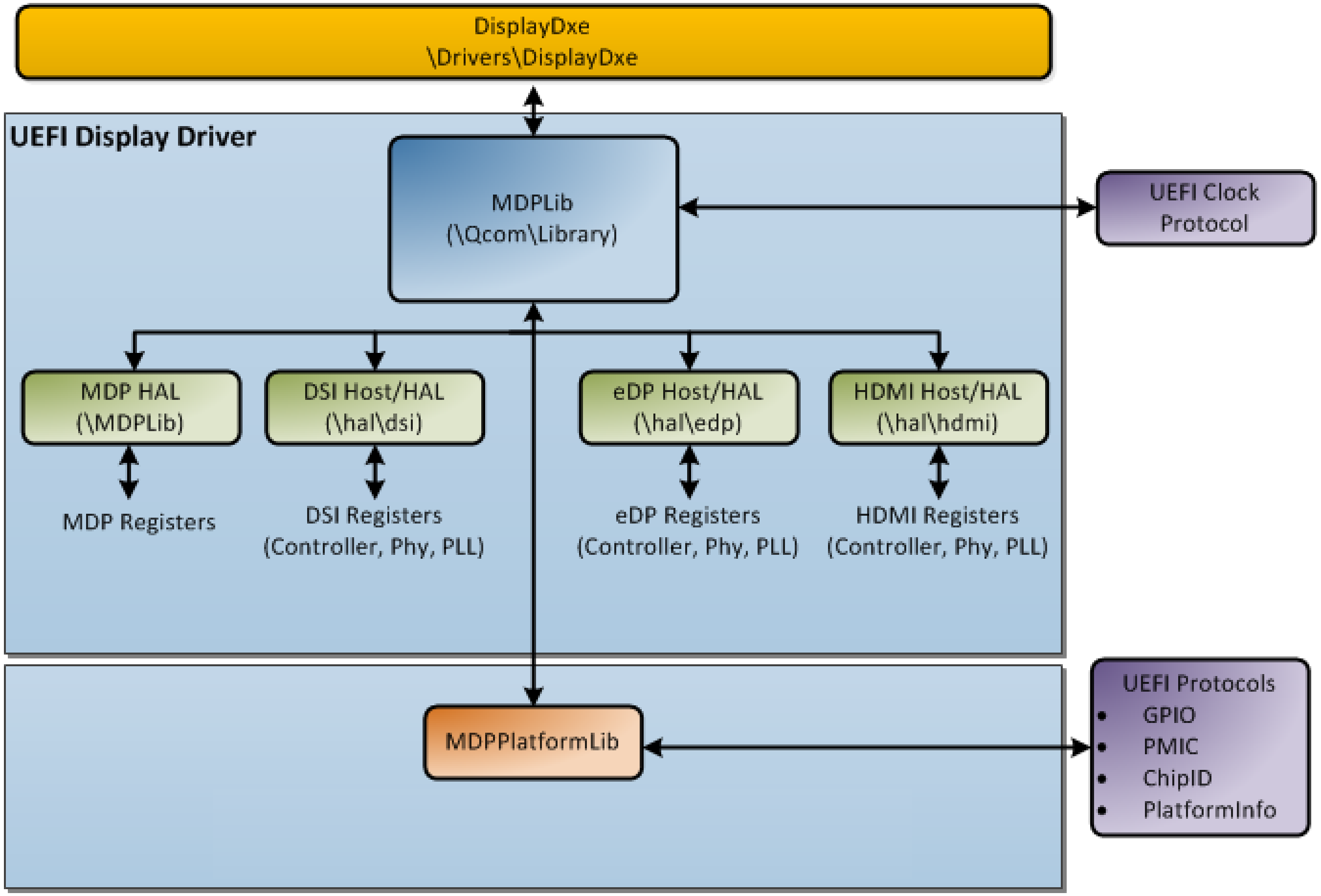
Display – UEFI

- UEFI
  - Replacement for BIOS in the desktop world or customized boot loaders in the mobile world.
  - The QTI reference version of UEFI is based on EDK2 (EFI’s open source implementation of UEFI).
  - QTI reference platform UEFI is recommended for licensee use.
  - A full source for QTI reference platforms is available to licensees.
- Display falls under the Display Dxe module category
  - All display code falls under a single Dxe module with dependencies on other system modules such as PMIC, GPIO, and clocks
  - QTI’s reference platforms implement the standard graphics output protocol (EFI_GRAPHICS_OUTPUT_PROTOCOL)

Graphics Output Protocol (GOP)

- Standard EFI protocol
  - EFI_GRAPHICS_OUTPUT_PROTOCOL_QUERY_MODE
    - QueryMode – Used to query the set of available display modes
  - EFI_GRAPHICS_OUTPUT_PROTOCOL_SET_MODE
    - SetMode – Used to set the display to a specific mode
  - EFI_GRAPHICS_OUTPUT_PROTOCOL_BLT
    - Blt – Used to bit application data to the screen
  - EFI_GRAPHICS_OUTPUT_PROTOCOL_MODE
    - Mode – Pointer to the current mode information

UEFI Driver Structure



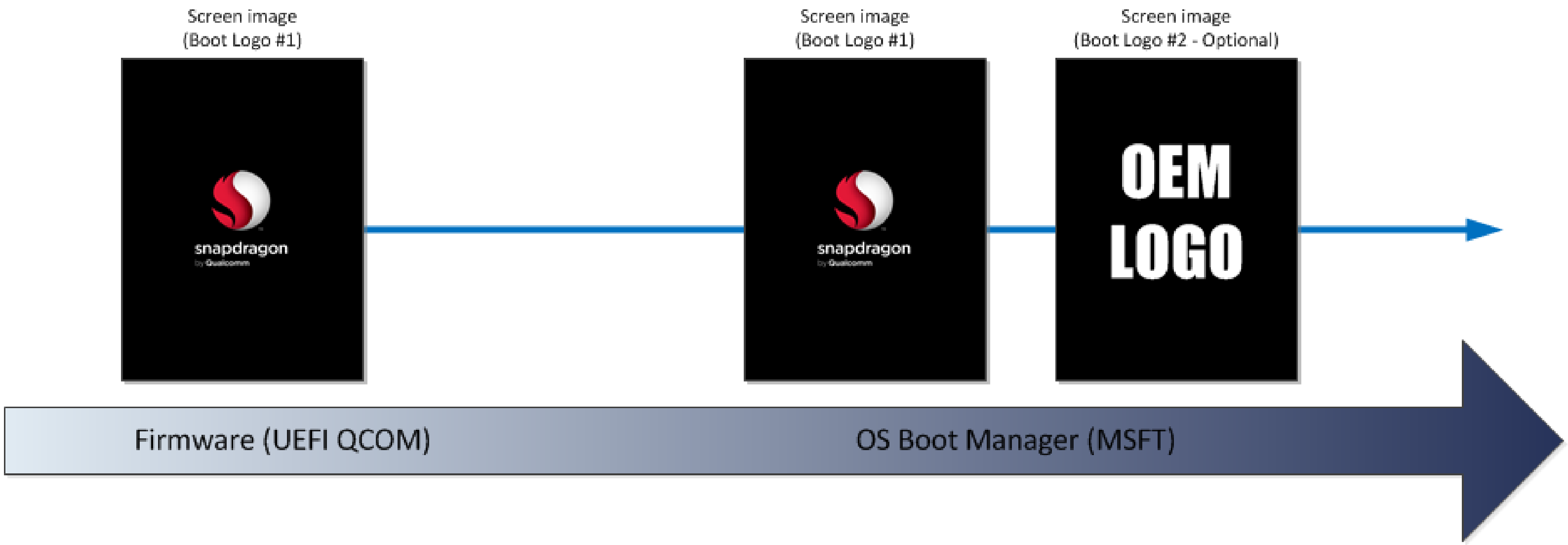
Section 7

WoS ACPI

Display – ACPI

- ACPI
  - Open standard for device configuration – windows relies heavily on ACPI to enumerate device driver, and provides configuration and resource information for all windows drivers.
  - Full source to ACPI tables are provided to licensees.
  - QTI reference supplies sample ACPI configuration to install and enumerate all resources for display, graphics, and video encoder/decoders.
  - ACPI also defines extension methods to allow licensees to customize their platforms.
    - None of the extensions are mandatory in Windows 8; it is up to the discretion of the miniport driver writer to use these extensions.
    - QTI provides samples for a few select display extensions. Most of the extensions are not supported by the QTI driver; however, licensees are free to implement and call ACPI methods in their own drivers.
    - Writing ACPI methods is even more limited than UEFI drivers; supporting complex drivers like DSI or HDMI in ACPI is impossible.

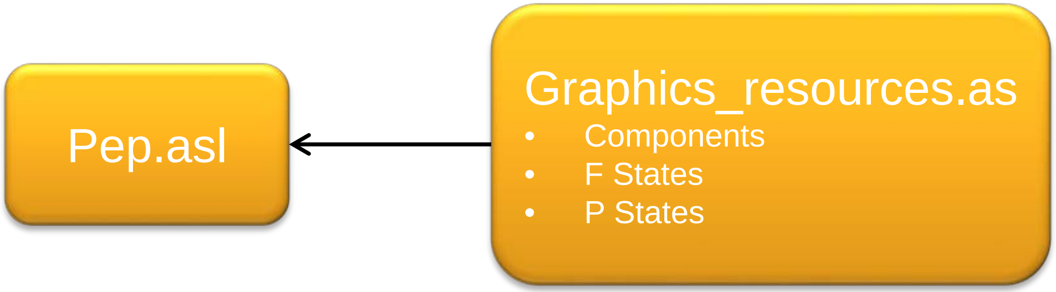
BGRT Table



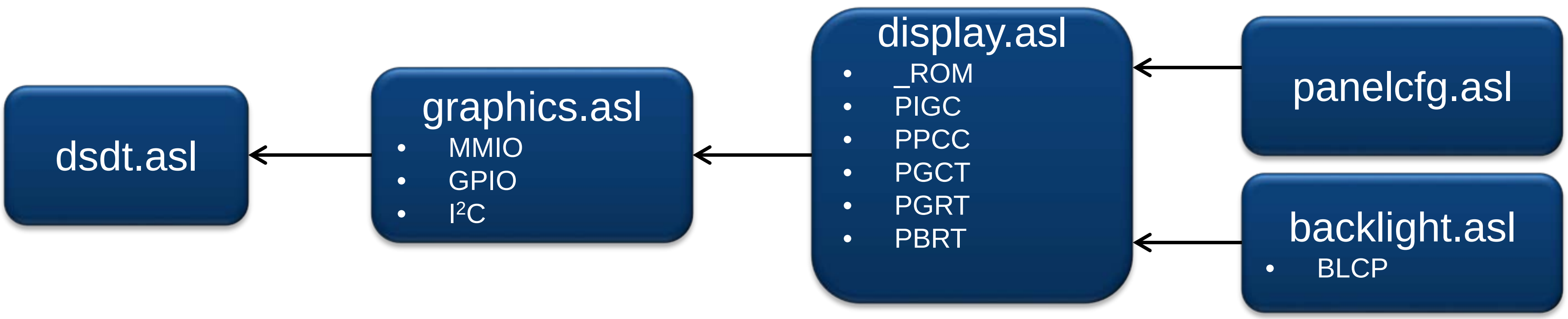
- The bitmap is displayed on screen by UEFI.
- A static BGRT table is used to communicate BMP information to the operating system.

ACPI Structure

ACPI – Power management



ACPI – Resources and configuration tables

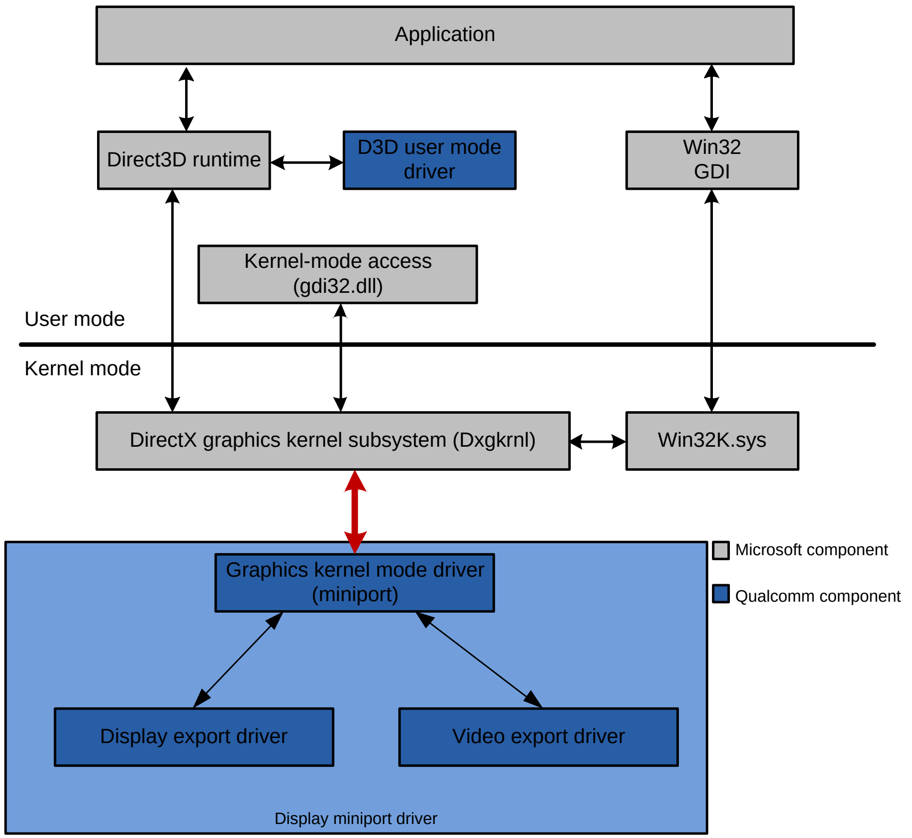


Section 8

WoS WDDM Model

WDDM Model

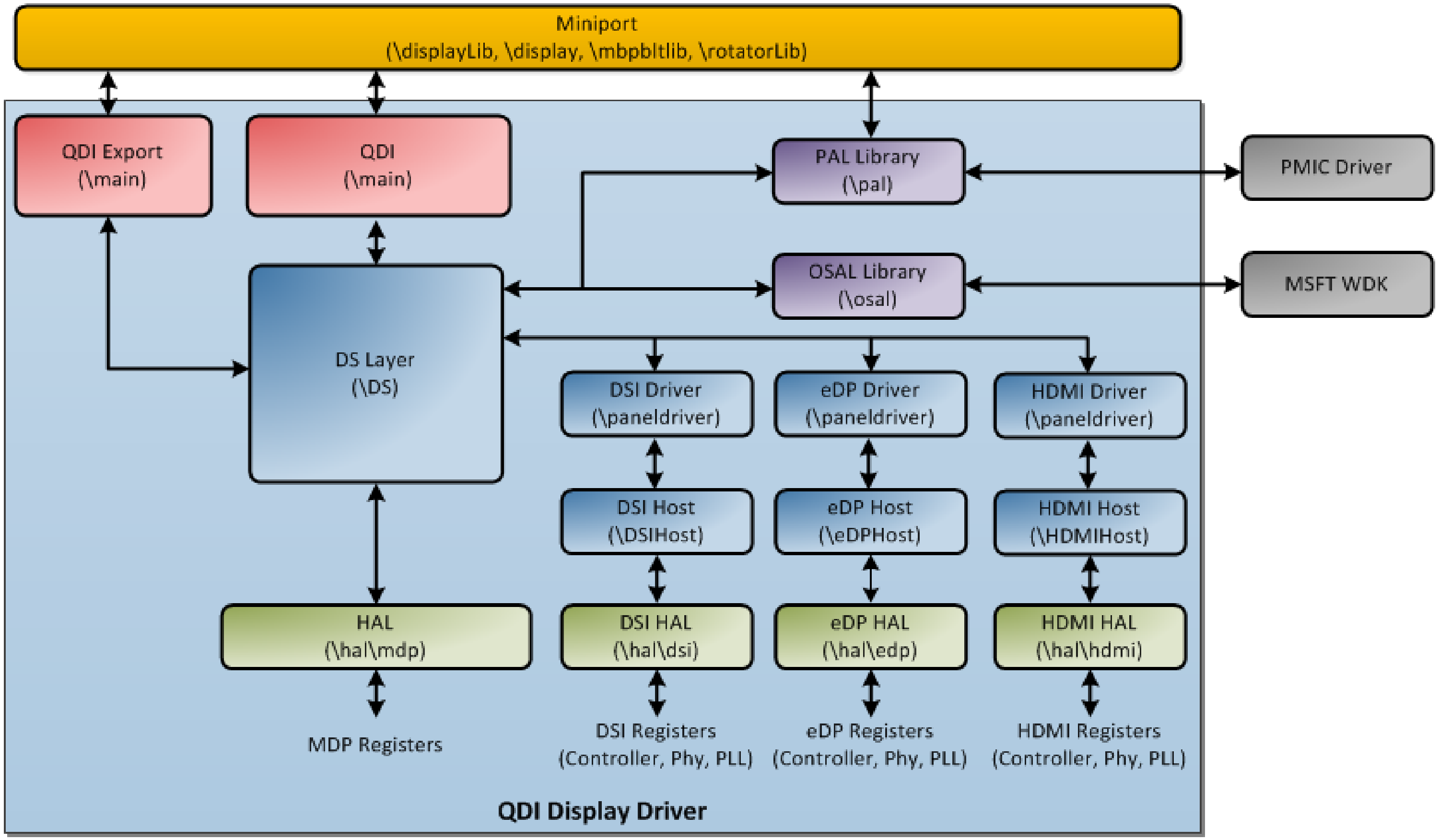
- Graphics kernel mode driver
  - Graphics driver and scheduler; direct interface to the Dxgkrnl DDI interface
- Display export driver
  - Display driver based on QDI interface; a separate driver is installed along with graphics KMD



WoS WDDM

- User mode components
  - D3D UMD
    - Manages all 3D rendering commands going to the GPU
  - D3D Video UMD
    - Manages all video commands to the video core and MDP bit engine
- Kernel mode components
  - Graphics KMD (also known as miniport)
    - Manages scheduling, surface allocation, GPU power management, source/target path, overlays, and hardware IP communication
    - Abstracts PEP/MSFT interfaces from QDI/Video engines and communication with other drivers
  - Display KMD
    - Hardware control of primary display, HDMI, overlays, bit operation, and rotation
  - Video KMD

QDI Driver Model



QDI Driver Modules

- QDI export
  - Special interface to the miniport
  - Interrupt handlers, DPC handlers, driver resource (MMIO, GPIO, and I^2C resources) registration.
- QDI
  - Standard QDI2 interface
  - Device, display, layer context handles, and management.
- PAL library
  - Platform abstraction layer
  - Miniport specific callbacks for ACPI manipulation, power management, GPIO, I^2C resource handling, and communication with PMIC (WLED and LPG)
- OSAL library
  - Operating system services abstraction, mutex, threads, delays, C Library (memcpy, memset), logs, and memory allocation.
- DS
  - Display services layer, display resource management, display, device, and layer control
  - Rotator, bit management
  - SMP management
- Panel drivers
  - Handle call sequencing and into eDP, DSI, and HDMI drivers
- Host drivers
  - Interface management of eDP, HDMI, and DSI protocols

Section 9

Windows Display Driver

Display Engine Concept

- Dxgkrnl manages display, graphics, and video kernel mode components as engines.
- Each engine is power managed separately and can be reset when it is nonresponsive (TDR).
- A TDR (timeout detection and recovery) is a method for windows to reset an engine if it does not respond to a command within an allowed timeout.
  - Display engine TDR – Occurs when the presented surface is not reported back during a Vsync interrupt due to following reasons:
    - Caused by lack of Vsync interrupt
    - Vsync interrupt took too long to occur
    - Surface (or surfaces) reported during a Vsync does not match the presented surface
  - Rotator/Bit TDR – Occurs when an operation fails to report a completion interrupt.
    - Hardware not reporting a complete interrupt as it goes into a loop or operation did not occur

Section 10

WoS Platform Customization

Display – Platform Customization (1 of 2)

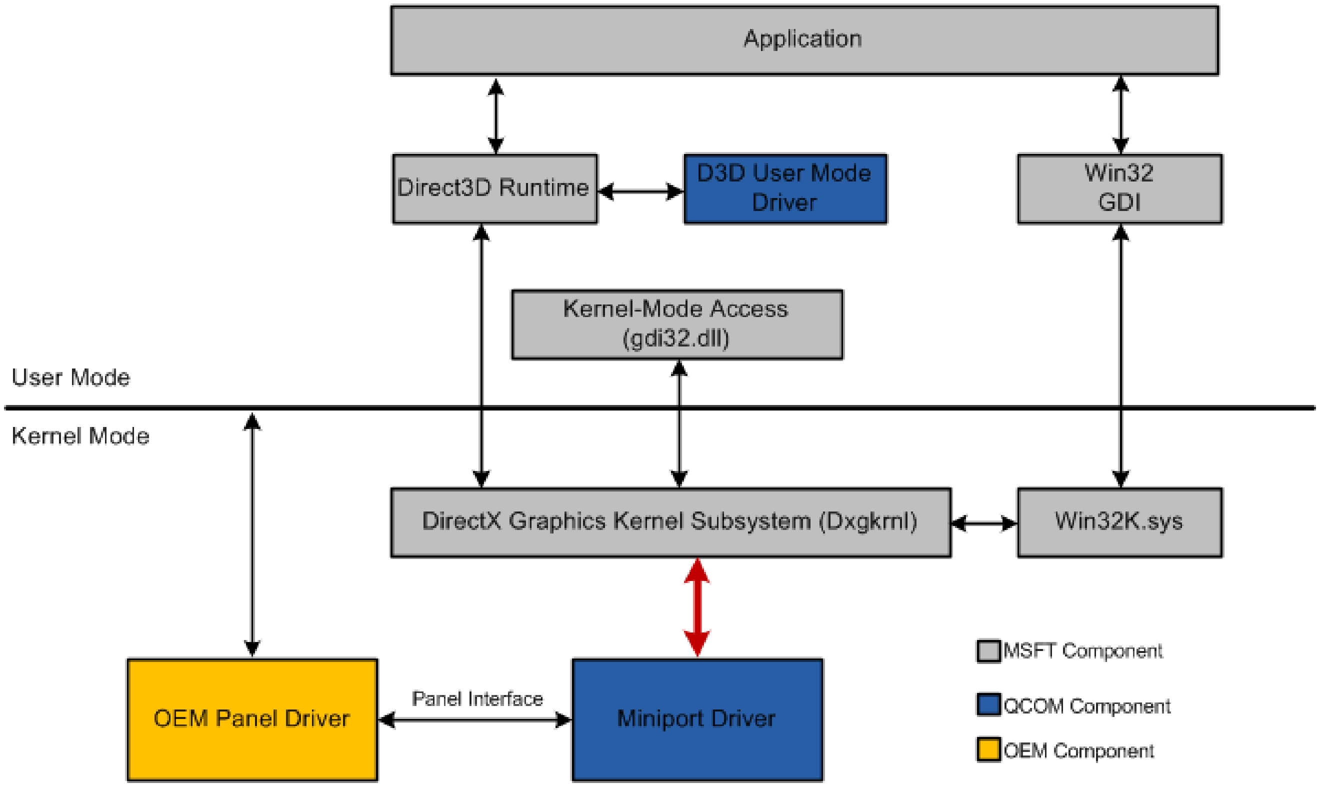
- Display customization for QTI reference platforms is performed entirely through the panel configuration file.
  - A human readable file (.xml-like) contains all the necessary fields and configuration to describe any active or smart panel
  - Panel configuration is split into several sections
    - EDID configuration – Windows requires a valid EDID to be presented for all internal and external panels; the panel .xml file requires manufacture ID, product ID, and other EDID-specific fields. Other fields (detailed timings) are simplified and retrieved from timing fields.
    - Timing fields – Active timing information, including active sizes, front/back porches, and pixel clock configuration.
    - Command fields – For DSI panels that support commands, the panel configuration file allows the licensee to send commands for specific events, such as panel initialization and panel shutdown.
    - Backlight control – Panels that require support backlight adjustment through PMIC or DSI commands are configured through the panel driver.

Display – Platform Customization (2 of 2)

- Panel configuration file customization
  - For UEFI, the panel driver is included in the UEFI code at compile time.
    - edk2\QcomPkg\Msm8974Pkg\Library\MDPPlatformLib\MDPPlatformLib.c
  - For the miniport, the panel driver is compiled into the ACPI tables via the _ROM method, which is a standard method to supply the miniport with proprietary information.
    - panelcfg.asl, backlightcfg.asl, display.asl, and pep.asl

Display – Advanced Platform Customization (Windows Phone 8 Only) (1 of 2)

- Although QTI reference platforms only use a panel configuration file, it is expected that OEMs require extra controls to fine-tune their platforms.
  - UEFI – Full UEFI source is available to OEMs; any customizations are done directly at a source level within UEFI.
  - Miniport – An OEM owned panel driver can be created to directly interface to the miniport driver.



Display – Advanced Platform Customization (Windows Phone 8 Only) (2 of 2)

- QTI provides a skeleton OEM panel driver
  - A published interface is defined to allow OEMs to override or customize a subset of miniport functions.
    - Oem_qdi.h (in QCDK)
    - OEM panel driver interface specification
  - OEMs own their respective panel drivers and do not need to publish this back to QTI. OEMs are required to supply a binary version of the OEM driver to debug platform-specific issues that cannot be reproduced without the OEM driver.
  - OEMs must abide by the OEM panel driver interface including rules for reentrancy, IRQL, and memory types (paged vs. nonpaged buffers).
- OEMs are responsible for installing their own panel driver, including any ACPI entries to facilitate driver installation.
  - Calling other drivers from the panel driver may require parent/child decencies or PnP notifications to be handled.
  - OEMs do not directly control the power states of the MDP hardware and interface cores. All power and performance controls are done through PEP ACPI entries (pep.asl).

Section 11

Gamma and Color Correction on WoS

Display Color Calibration

- Panels inherently have different behaviors between manufactures, models, and even between individual panels.
- Involves four components:

1. Modify panel gamma LUT to correct **panel tone**.
   - Based on the panel LUT table gathered from display calibration to define matching gamma curve (MGC).
2. Modify display CSC block to correct **panel color**.
   - Modify 3×3/3×11 matrix to adjust panel response based on data from display calibration.
3. Modify gamma LUT to correct **picture gamma**.
   - Apply gamma correction on a picture based on a set of predefined gamma curves (REC601, REC709, and so on)
4. Modify display CSC block to **dynamically adjust** hue, saturation, and intensity of the content.
   - Adjust 3×3/3×11 matrix to manipulate hue, saturation, and intensity based on application control.
   - Same feature can be applied at the display pipe as well as VG layer pipes.

ACPI Entries for Panel Calibration

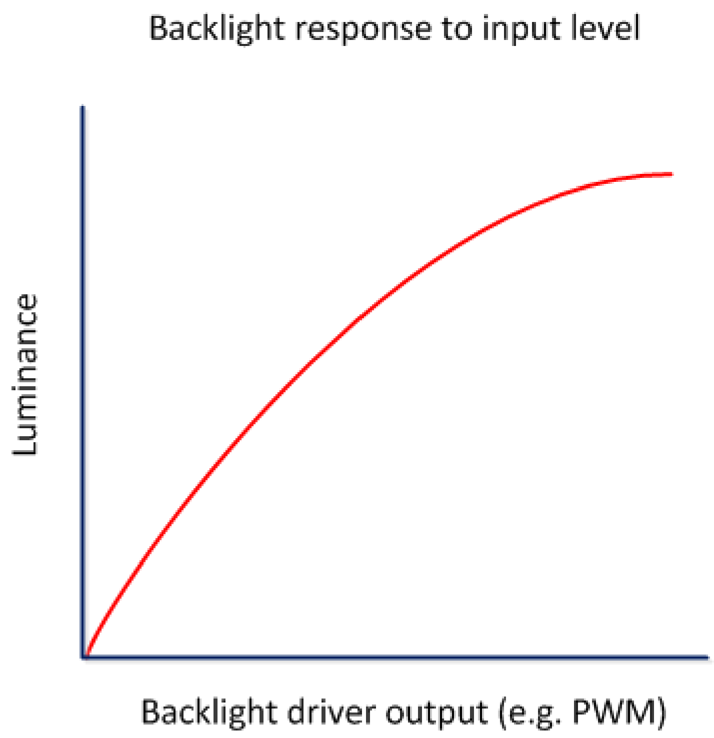
- The panel calibration data is stored in the ACPI table, and the data is read out during the panel initialization, and then get programmed into hardware blocks.
- There are three ACPI methods defined for driver to retrieve the corresponding data:
  - PIGC – To retrieve panel inverse gamma correction data (3 × 256 256 × 16 bits)
  - PPCC – To retrieve panel color correction data (3 × 11 11 × 64 bits)
  - PPGC – To retrieve panel gamma correction data (3 × 16 16 × 32 bits)

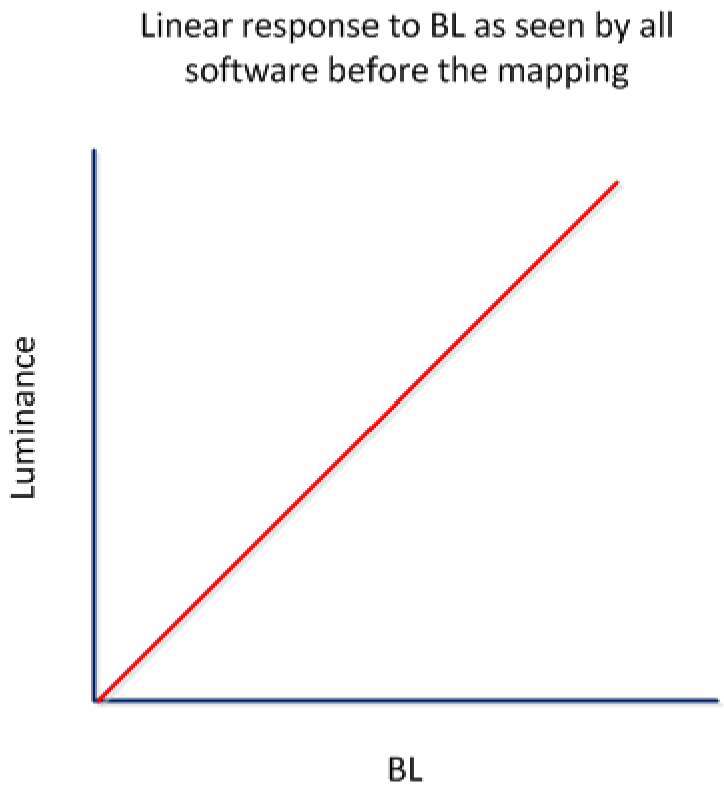
Color Management Blocks

- PIGC
  - Inverse gamma table
- PGRT
  - Gamma response table
- PPCC
  - Polynomial color correction table
- PGCT
  - Gamma correction table
- PBRT
  - Backlight response table

Panel Backlight Response

- Panels may not have a linear response to the backlight level applied to them.
- Entry measurement is on PBRT to linearize the curve.





Backlight Response Mapping Overview

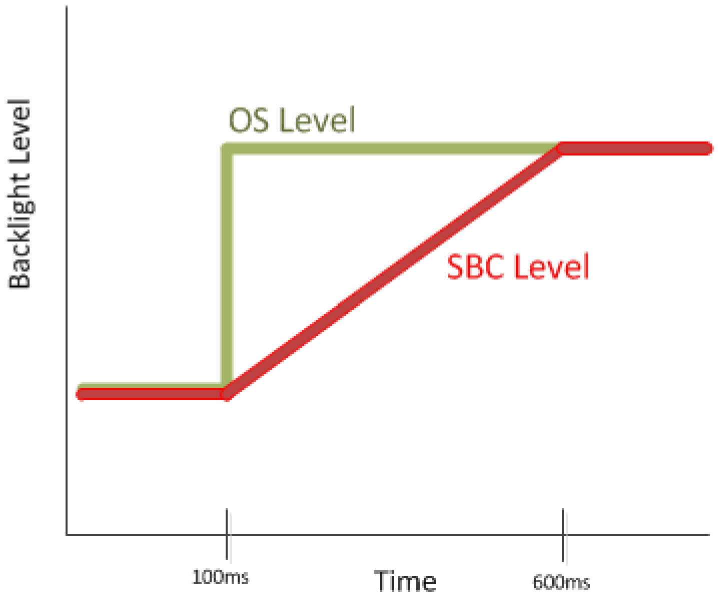
- Backlight response mapping serves as a calibration of backlight response so that correct luminance is achieved.
- The OEM calibrates panels and stores the response table in ACPI.
- Calibration involves an accurate measurement of luminance.
- A display driver uses this table to do the mapping as the last stage of backlight calculations.

Section 12

Backlight and Image Optimizations

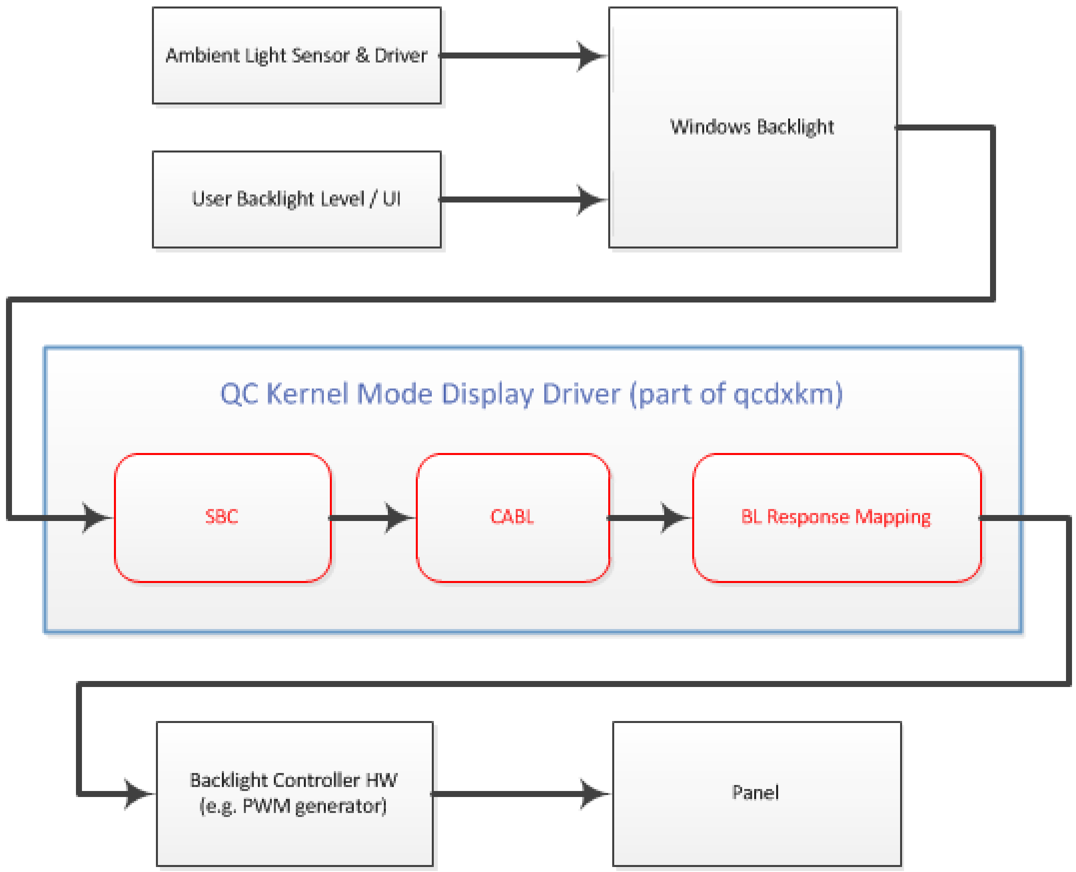
Smooth Backlight Control (SBC)

- SBC is a feature to make backlight changes smooth instead of drastic or instantaneous.
- Applies to all backlight changes made by the user or the operating system, except when turning off backlight (needs to be immediate).



Content Adaptive Backlight Leveling (CABL)

System-wide processing of backlight (except assertive display)



Overview of CABL

Algorithm to reduce power by analyzing the frame contents.

- Backlit (LCD) displays
  - Algorithm reduces backlight level
  - Algorithm compensates by boosting pixel values using a LUT
  - Algorithm pursues maintaining image fidelity, but it could be degraded
- AMOLED displays
  - Displays without backlight, each pixel emits its own light
  - Algorithm reduces pixel values
  - Algorithm enhances contrast
  - Image fidelity loss is unavoidable (may not be perceptible)

Windows Control of CABL

Windows operating system controls dynamically the following parameters:

- CABL enable/disable
  - Enabled when using battery power
  - Disabled when plugging a power adapter
- CABL power savings setting based on use case
  - UI/Desktop – low-power savings
  - Full screen video – high-power savings
  - Dimmed – medium power savings

Section 13

MDP Extensions in Qualcomm® Adreno™ OEM Debugger Extension

MDP Extensions in Adreno OEM Debugger Extension

!mdp_chipid	Displays the MDP chip ID
!mdp_int_status	Returns MDP enabled interrupts and status with an option to clear
!mdp_mixer_info	Returns the MDP layer/mixer routing and hardware status
!mdp_layer_info	Returns the MDP layer configuration information
!mdp_axi_info	Returns the MDP AXI routing and status
!mdp_blend_info	Returns the MDP blending configuration
!mdp_cursor_info	Returns the MDP cursor configuration
!mdp_status	Returns the MDP hardware status
!mdp_bist	Enables/disables the MDP built-in-self test
!mdp_reg_base	Configures the base address for mdp_rr and mdp_wr
!mdp_rr	Read the MDP register at offset
!mdp_wr	Write the MDP register at offset with value
!mdp_dump_regs	Dump all the MDP registers to file
!mdp_screencapture	Perform a MMU-based screenshot of the frame buffer

Questions?

For additional information or to submit technical questions, go to:
https://discuss.96boards.org/c/products/dragonboard410c

EXHIBIT 1

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2. **COMPLIANCE WITH LAWS; APPLICABLE LAW.**

Any litigation or other dispute resolution between You and Us arising out of or relating to this Agreement, or Your relationship with Us will take place in the Southern District of California, and You and QTI hereby consent to the personal jurisdiction of and exclusive venue in the state and federal courts within that District with respect any such litigation or dispute resolution. This Agreement will be governed by and construed in accordance with the laws of the United States and the State of California, except that body of California law concerning conflicts of law. This Agreement shall not be governed by the United Nations Convention on Contracts for the International Sale of Goods, the application of which is expressly excluded.

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